

# ELIAN HIJMANS MALESSY

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## RESEARCH INTERESTS

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Probabilistic Models, Generalization Theory, Bayesian Deep Learning, AI For Science

## EDUCATION

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### University of Amsterdam (UvA)

M.S. in Artificial Intelligence

Sept. 2026 – Present

Amsterdam, Netherlands

### University of California, Irvine

B.S. in Computer Science, Specialization in Intelligent Systems

GPA: 3.93 / 4.00; *Cum Laude*; *Dean's Honor List* (all quarters)

Sept. 2023 – March 2026

Irvine, California, USA

## RESEARCH & RELEVANT EXPERIENCE

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### Uber — Cassandra Database Team

Software Engineer Intern

Jun. 2026 – Aug. 2026

Sunnyvale, CA

- Modeled cluster memory dynamics using historical heap metrics across the cluster, allowing services to predict how the heap size will change after adjusting node parameters and changes in traffic levels.
- Engineered an automated heap scaling service for Uber's Cassandra fleet, reducing total fleet production memory allocation by 51%.

### UC Irvine — Intelligent Dynamics Lab

Undergraduate Researcher (Advisor: Roy Fox)

Apr. 2025 – Mar. 2026

Irvine, CA

- Developed a PyTorch-based library for stochastic optimization in reinforcement learning, implementing gradient estimators and control variates with JIT-compiled modules to accelerate training.
- Improved policy learning in DreamerV3 by replacing straight-through estimators with Gumbel-Softmax relaxations for discrete latents and actions, enabling stable backpropagation through time in discrete environments.

### UC Irvine — Machine Learning Research Group

Undergraduate Researcher (Advisor: Alexander Ihler)

Apr. 2025 – Mar. 2026

Irvine, CA

- Investigated the double descent phenomenon through empirically evaluating how model scale, distillation, and regularization influence minima flatness and generalization.
- Merged pruned CNNs into a parameter-efficient multi-head model by combining similar filters and running unique filters through grouped convolutions.

### Microsoft — Surface Camera Software Team

Software Engineer Intern

Jun. 2025 – Sep. 2025

Redmond, WA

- Developed thermal throttling functionality for Surface camera drivers to optimize performance under AI workloads, strengthening skills in systems programming and performance optimization.
- Built an automated camera sensor validation pipeline and optimized Hardware Lab Kit workflows, reducing driver smoke test runtime from days to under one hour.

### California Institute for Telecommunications and Information Technology

Research Assistant / Software Developer

Sep. 2023 – Apr. 2025

Irvine, CA

- Developed physics-informed neural networks to serve as surrogates for finite-element thermal simulations, reducing computation time for energy evaluations in thermal-storage systems.
- Built real-time video analysis pipelines using pretrained CNNs for detection, tracking, and pose estimation on security camera feeds, part of a proof of concept security system.

## SELECTED PROJECTS

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### Molecular Generation

Feb. 2026 – Mar. 2026

- Implemented and trained a Graph Variational Autoencoder on the ZINC-250k dataset to generate drug-like molecules.
- The final architecture includes a latent property prediction head, an inverse autoregressive flow, and an autoregressive transformer decoder, achieving a Fréchet ChemNet Distance of 1.81.

### Semi-Supervised Programming Language Translation

Feb. 2026 – Mar. 2026

- Fine-Tuned a CodeT5 model to be able to translate between Python and C++.
- Trained with a semi-supervised approach consisting of supervised learning on code snippet pairs, and unsupervised learning using backtranslation and denoising autoencoding on an unpaired dataset.
- Fine-tuning CodeT5-base resulted in a CodeBLEU score of  $\sim 76\%$ , and a HumanEval-X score of  $\sim 20\%$ .

## TECHNICAL SKILLS

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**Languages:** Python, C++, C, Go, Java, JavaScript, SQL

**ML & Data Science:** PyTorch, JAX, NumPy, Pandas, Scikit-learn, OpenCV, Apache Spark

**Tools & Platforms:** Apache Cassandra, Git, Docker, Slurm, Apache Kafka, AWS, PostgreSQL, Flask

## RELEVANT COURSEWORK

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**Graduate:** Deep Generative Models

**Machine Learning & AI:** Machine Learning and Data-Mining, Neural Networks and Deep Learning, Algorithms for Probabilistic and Deterministic Graphical Models, Applications of Probability in Computer Science

**Math:** Statistics, Linear Algebra, Multivariable Calculus, Differential Equations

## REFERENCES

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**Alexander Ihler, Ph.D.** — Professor, Department of Computer Science, UC Irvine — [ihler@ics.uci.edu](mailto:ihler@ics.uci.edu)

**Roy Fox, Ph.D.** — Assistant Professor, Department of Computer Science, UC Irvine — [royf@uci.edu](mailto:royf@uci.edu)

**Sesha Sailendra Chetlur** — Senior Software Engineer, Surface Software Team, Microsoft — [schetlur@microsoft.com](mailto:schetlur@microsoft.com)